Journal of Physics G: Nuclear and Particle Physics

PAPER

# How informative is agreement among nuclear-mass models? Chronology-respecting tests of error coverage and shell proximity

Othman H. Y. Zalloum[1,*]

[1]Department of Applied Mathematics and Physics,
Palestine Polytechnic University,
Hebron, West Bank, Palestine
*Corresponding author: Othman H. Y. Zalloum

**E-mail:** ozalloum@ppu.edu

**Keywords:** nuclear masses; model disagreement; uncertainty calibration; AME2020; shell structure; two-neutron separation energy

## Abstract

The spread of predictions from different nuclear-mass models is often displayed as a measure of disagreement, but its relation to subsequent error must be measured rather than assumed. We study the equally weighted predictions of FRDM2012, HFB-27, DZ31, and WS4 using nuclei newly represented by experimentally evaluated masses in successive Atomic Mass Evaluations (AMEs). An AME2012–2016 cohort ($n = 32$) defines the exploratory analysis, followed by an AME2016–2020 cohort ($n = 51$, experimental mass uncertainty below 100 keV) and a prespecified broader confirmation cohort ($n = 71$) without that uncertainty threshold. Let $D$ be the sample standard deviation of four mass predictions and $E$ the absolute error of their mean. The Spearman association is positive in both confirmation cohorts: $\rho = 0.265$ (nominal $p = 0.060$) in the primary 51 and $\rho = 0.329$ (nominal $p = 0.005$) in the broad 71. Empirical coverage of the mean $\pm D$ is 42/51 (82%) and 57/71 (80%); for $\pm 2D$ it is 49/51 (96%) and 68/71 (96%). The ratio of RMS error to RMS spread is 1.25 in development, 0.70 in primary confirmation, and 0.82 in broad confirmation. In the primary confirmation cohort, the nine nuclei near specified shell closures have MAE 0.593 MeV versus 0.173 MeV elsewhere. These historical samples are clustered and span only short extrapolation distances; they do not support statements about deep neutron-rich extrapolation. Raw inter-model spread has useful aggregate association and coverage in this restricted regime but no model-independent probabilistic meaning; its numerical mapping to errors and its behavior near shells require validation for each application.

## 1 Introduction

Global nuclear-mass formulas are used where measurements are not available, for example in studies of nuclear structure and astrophysical nucleosynthesis. Different formulas may give similar numbers. Such agreement is sometimes shown as a theoretical spread or systematic model-dependence band, but a small spread alone does not tell us how far the shared prediction is from nature. The distinction is particularly important when all models share mass-systematics information or omit similar local shell physics [5, 7, 20].

Uncertainty studies in nuclear physics already go far beyond naive model spread. Earlier tests compared global predictions against subsequent nuclear masses [16]; Bayesian Gaussian-process and neural-network corrections were evaluated for new separation-energy data using empirical coverage curves [17]; and an ensemble Bayesian-model-averaging scheme tested probabilistic uncertainty estimates for one-neutron separation energies from AME2003 against AME2020 [6]. Broader treatments explain statistical model discrepancy and nuclear-theory uncertainty [18, 19]. Model orthogonalization explicitly addresses the risk of counting highly correlated calculations as independent evidence [20]. These studies are important precedents, not methods that the present calculation supersedes.

For example, Utama and Piekarewicz [16] examined 46 new AME2016 masses in a restricted medium/heavy-mass region after Bayesian neural-network refinement of global models. Our

**Table 1.** Chronology-defined populations. Primary requires experimental mass-excess uncertainty below 100 keV; broad removes only that condition. $S_{2n}$ counts require a non-estimated same-evaluation neighbor.

| Population | $n$ |
|---|---|
| AME2012 / AME2016 / AME2020 primary measured | 2235 / 2271 / 2339 |
| 2012–2016 development, primary / broad | 32 / 58 |
| 2016–2020 confirmation, primary / broad | 51 / 71 |
| Development with $S_{2n}$, primary / broad | 17 / 41 |
| Confirmation with $S_{2n}$, primary / broad | 34 / 44 |

development population has 32 nuclei because we require a new non-estimated entry, $N, Z \geq 8$, experimental uncertainty below 100 keV, and complete predictions from *all four* fixed global models; our observable is the spread of their *uncorrected* predictions. The two validation problems should not be conflated. Similarly, the probabilistic coverage analyses of Neufcourt *et al.* and Saito *et al.* evaluated explicitly fitted uncertainty distributions, whereas here there is no posterior or learned model discrepancy to begin with [6, 17].

Our narrower question concerns an unmodified, equally weighted collection of four heterogeneous global models: the macroscopic–microscopic FRDM2012 [1], the density-functional HFB-27 [2], the Duflo–Zuker DZ31 calculation [3, 15], and the Weizsäcker–Skyrme WS4 formula [4]. We do not infer mixture weights, train an error corrector, or use the later outcomes to select models. WS4's reported accuracy relative to several alternative formulas [8] also means that this fixed equal-weight mean is a diagnostic of raw agreement, not a proposed optimal predictor.

Three statistical questions need to be kept separate. *Accuracy* asks how close the center of the four predictions is to experiment. *Error discrimination* asks whether larger disagreement tends to accompany larger errors across nuclei. *Uncertainty calibration* asks whether intervals constructed from the disagreement have interpretable numerical coverage, and whether that mapping transfers to later samples. Rank correlation addresses the second question, not the third.

We use experimentally evaluated outcomes first appearing in AME2016 and AME2020 relative to the immediately preceding evaluation. Development-defined diagnostics from the first transition are applied without adjusting their thresholds to the next transition. We additionally report the prespecified population without the 100 keV experimental-uncertainty exclusion, block resampling by isotopic chain, direct interval coverage, scale ratios, and a predeclared calcium-shell discussion. We deliberately refer to this design as *chronology-respecting with an analysis plan*: a document in a public release dated after the calculations is not evidence of externally witnessed preregistration, and experimental AME chronology is not identical to each model's fitting chronology.

## 2 Data and methods

### *2.1 Atomic-mass chronology*

We use the mass-excess tables of AME2012, AME2016, and AME2020 [9–14]. A new experimental outcome is a nucleus that lacks a finite, non-estimated mass excess in one evaluation but has one in the next; estimates marked with **#** and unavailable entries are excluded. Both proton and neutron numbers must be at least eight. The *primary* population additionally requires mass-excess uncertainty below 100 keV. In the *broad* sensitivity population this last cut is removed, but estimated masses remain excluded. The broad comparison includes the 20 additional AME2016–2020 nuclei whose mass uncertainties are 102–420 keV, rather than silently dropping them from the scientific discussion.

The transition AME2012→AME2016 is the development cohort (32 primary, 58 broad). AME2016→AME2020 is the confirmation cohort (51 primary, 71 broad). For reproducibility, the numerical results here were independently reconstructed from the publicly released AME membership and standardized model tables. In a subsequent source audit we independently parsed the three official fixed-width AME tables, recovering all 3353, 3436 and 3558 archived entries, their mass excesses and their eligibility flags, respectively. This checks the raw-to-standardized AME outputs without relying on the absent historical parser code (see Data Availability). The first cohort fixes all thresholds, and none is refitted after viewing confirmation errors.

### *2.2 Consensus and uncertainty signal*

For model prediction $M_{ik}$ of nucleus $i$,

$$\bar{M}_i = \frac{1}{4}\sum_{k=1}^{4} M_{ik}, \quad D_i = \left[\frac{1}{3}\sum_{k=1}^{4}(M_{ik} - \bar{M}_i)^2\right]^{1/2}, \tag{1}$$

and $E_i = |\bar{M}_i - M_i^{\rm exp}|$. Complete four-model coverage is required. The Spearman statistic $\rho(D, E)$ measures whether spread ranks nuclei by their later errors. It is a measure of *discrimination*, not numerical calibration. We therefore also report empirical coverages $C(c) = n^{-1} \sum_i \mathbf{1}\{E_i \le cD_i\}$ for $c = 1, 2$, Wilson intervals as per-nucleus descriptive bounds, and the descriptive scale ratio

$$k_{\rm RMS} = \left[\frac{\sum_i E_i^2}{\sum_i D_i^2}\right]^{1/2}. \tag{2}$$

This is the ratio RMS(error)/RMS(spread), not an assumed 68% predictive standard deviation. We give the scale separately for development and each confirmation population, with ordinary and isotopic-chain bootstrap intervals. For illustration only, four independent, identically distributed, zero-mean Gaussian model errors would give $E/(D/2) \sim |t_3|$, predicting approximately 86% and 97% coverage for $\pm D$ and $\pm 2D$. Real global mass models are neither independent nor exchangeable; these percentages are a benchmark, not a justification for assigning such probabilities to $D$.

Development-defined $D$ quintile edges are reused exactly in confirmation; each bin reports its count, mean error and 68%/95% percentile bootstrap intervals from 10,000 resamples (seed 42). We also compare binned RMS error with binned RMS spread. The interval analysis remains descriptive because outcome dependence means that resampling individual nuclei can exaggerate precision. Thus, every headline association and scale ratio is also assessed by resampling whole isotopic chains (the $Z$ blocks) with replacement. The same 10,000-draw, seed-42 $Z$-block percentile bootstrap is used for mean absolute error (MAE), empirical $C(D)$ and $C(2D)$, and the near-minus-other shell-proximity MAE contrast, so every principal aggregate outcome has a clustering-aware 95% interval. Within each resample the denominator is the total number of sampled nuclei, including all observations in each selected chain. The RMS scale-ratio intervals use 10,000 draws with seed 53, as documented in Supplement S1. Reported Spearman $p$-values are ordinary asymptotic two-sided values; they are not adjusted for isotopic-chain dependence or for multiple comparisons. The primary and broad cohorts overlap and are not independent replications. Interpretation therefore emphasizes effect sizes, the chain-bootstrap intervals and the prespecified scope of each analysis. Resampling $Z$ blocks preserves dependence within a chain but assumes independence across chains; it does not remove broader regional dependence.

As a supplementary proper-score check, we form a transparent working Normal predictive distribution with center $\bar{M}_i$ and scale $\sigma_i = k_{\rm RMS,\,dev} D_i$, where the scale is learned from the *primary development cohort only*. We evaluate its continuous ranked probability score (CRPS) on confirmation, alongside a constant-scale Normal baseline with $\sigma_0 = {\rm RMS}(\bar{M}_i - M_i^{\rm exp})$ fitted on the same development cohort. These are deliberately simple scoring comparators, not probabilistic outputs supplied by any of the mass models.

### *2.3 False consensus and sensitivities*

Two low-spread, large-error events are evaluated using fixed definitions. The stringent absolute event FC-A requires $D \le 0.25$ MeV and $E \ge 1.00$ MeV. FC-B requires $D \le 0.238$ MeV and $E \ge 0.341$ MeV, the original development lower-spread and upper-error quartiles (the unrounded values are used in software). The latter is a sensitivity test, not a universal physical threshold. We report counts and Wilson 95% intervals with clear denominators. A zero count is considered informative only if the cohort is capable of meeting both parts of the proposed event. To test dependence on ensemble membership without choosing a model after the fact, each of the four three-model leave-one-out means is also evaluated, including its own spread–error association and coverage. Pairwise signed-residual correlations describe shared model errors but do not imply a causal mechanism.

### *2.4 Nuclear-chart conditions and statistical power*

For each target, $d_2$ is its smallest Euclidean distance in the $(N, Z)$ lattice to a primary measured nucleus in the preceding AME; $d_1$ is the corresponding Manhattan distance. The shell-proximity flag is one when $N$ lies within two units of 20, 28, 50, 82 or 126, or $Z$ within two units of 20, 28, 50 or 82. Magic number 8 was not in the specified list; adding it does not change either confirmation population, as none of its targets has $N$ or $Z$ between 8 and 10. We report the flag, the continuous distance associations and isospin asymmetry $(N - Z)/(N + Z)$.

The prespecified development-based distance terciles have a technical limitation: repeated nearest-neighbor distances of exactly one lead to coincident thresholds, so the middle bin contains no confirmation nuclei. We disclose this and also report the continuous $d_1, d_2$ associations, instead of silently presenting the two populated bins as if they formed three equal-sized groups. With 51 nominally independent nuclei, a two-sided 5% test has approximately 47% power to detect a true

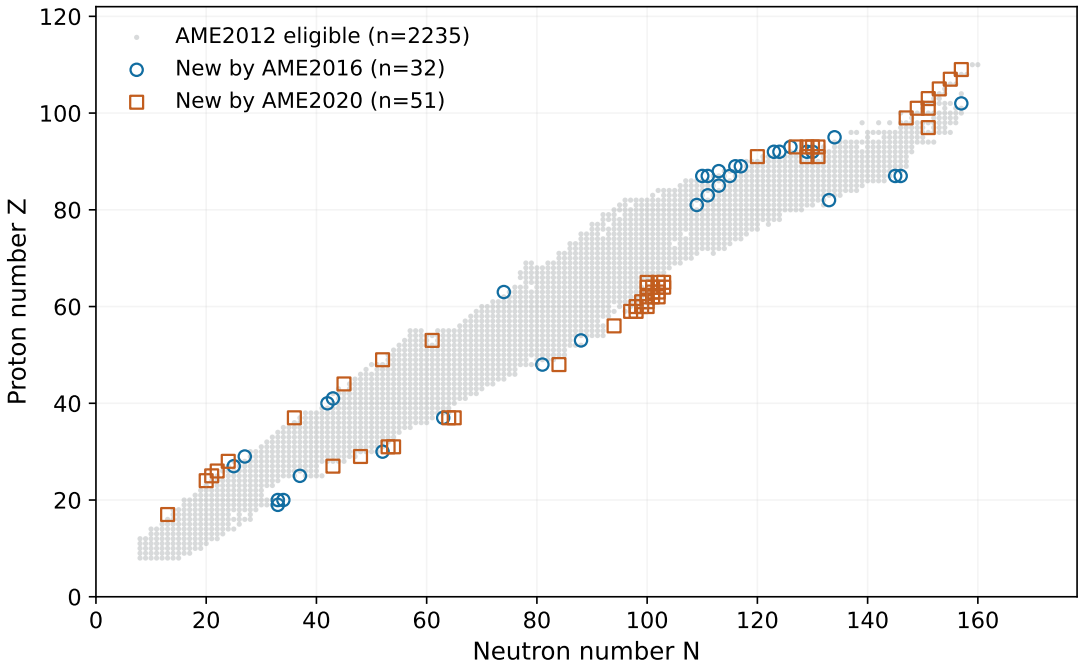



**Figure 1.** Experimental mass-evaluation chronology on the $(N, Z)$ chart. Gray points represent the 2235 primary eligible AME2012 masses; open blue circles identify 32 newly eligible AME2016 targets, and open orange squares identify 51 newly eligible AME2020 targets. Here "confirmation" means a later experimental evaluation, not necessarily exclusion from every theoretical model's historical fit.

correlation of 0.265 (Fisher-$z$ approximation); roughly 110 independent targets would be required for 80% power at that effect size. This is a rough independent-data planning approximation applied to a rank correlation, not an exact Spearman power calculation or additional evidence from the observed $p$-value. Isotopic-chain dependence further reduces effective information.

### *2.5 Structural secondary endpoint*

The secondary observable is

$$S_{2n}(N, Z) = ME(N - 2, Z) + 2\Delta_n - ME(N, Z), \tag{3}$$

with $\Delta_n = 8.0713181$ MeV. Each model uses its own target and neighbor masses; experimental values require a measured, non-estimated neighbor in the later AME.

The target and its $N - 2$ neighbor must both have non-estimated experimental mass excesses in the later evaluation; the latter need not meet the primary target's 100 keV cutoff. Each theoretical $S_{2n}$ is calculated within one model, avoiding cross-model mixing. Its four-model spread and consensus error are defined exactly as for mass excess. This is a secondary diagnostic of local mass structure, not an independent validation sample because its underlying masses and neighboring residuals are correlated.

## 3 Model provenance and validation

We use only four identified, fixed prediction tables. Model-prediction data are mass excesses in MeV keyed uniquely by $(N, Z)$, with coverage and basic cross-source identity tests stated below. Prediction-table size is not the number of fitted masses. The approximate fitting chronology differs by model: FRDM2012 adjusted ten macroscopic constants to 2149 then-available masses, HFB-27 fitted essentially all 2353 masses available at its 2013 publication, and WS4's 2014 parameters reflect the then-available mass database [1, 2, 4]. For the precise DZ31 program used here, the release does not supply an independently auditable fitting-mass list or its adjustment date. It must not be assumed that all models share one training vintage. This asymmetry affects disagreement $D$: part of a model difference may reflect fit age as well as extrapolation difficulty.

FRDM2012 and WS4 standardized tables originate in an IAEA-NDS RIPL-4 distribution, HFB-27 in BRUSLIB, and the DZ31 predictions in the AMDC Duflo–Zuker 31-parameter software [15]. The release reports independent table agreement on 8539 common FRDM2012 nuclei and agreement of BRUSLIB HFB-27 with the RIPL-4 copy on 8386 nuclei within 5 keV of printed precision. Our independent audit additionally reproduces all 9318 FRDM2012 and 10237 WS4

**Table 2.** Fixed model ensemble. "Predicted rows" is the number in the standardized prediction file; it is *not* a fit-set size. A dash means the exact fit population could not be verified from the source materials.

| Model | Predicted rows | Fit masses | Source and fit vintage |
|---|---|---|---|
| FRDM2012 | 9,318 | 2,149 | RIPL-4; published 2016 |
| HFB-27 | 8,386 | 2,353 | BRUSLIB; fitted by 2013 |
| DZ31 | 21,275 | — | AMDC 31-parameter program; fit set unverified |
| WS4 | 10,237 | — | RIPL-4; 2014 formula; exact fit set unverified |

mass-excess entries exactly from the official RIPL-4 inputs. The 8386 archived HFB-27 entries match the independent, more precisely printed RIPL-4 distribution to within 5 keV; this is an independent identity check, not recovery of the original BRUSLIB-to-CSV transformation.

The original Duflo–Zuker paper reports a 28-parameter fit, so it must not be used alone as documentation of the later AMDC 31-parameter executable [3, 15]. The recovered official `dz31.f` program enumerates exactly the 21,275 $(N, Z)$ points present in the archive. It was compiled and executed on a Windows host with GNU Fortran 16.0.1 (`-std=legacy`, `-ffixed-line-length-none`, `-O0`). An independently written Fortran checker compared all 21,275 printed mass-excess predictions with the archived DZ31 table by $(N, Z)$: every key and mass number matched, and the maximum absolute discrepancy was 0.000 MeV at the tables' 0.01-MeV reporting precision (acceptance threshold 0.011 MeV). The original 957,375-byte `massex.out` has also been recovered from an earlier revision archive and is distributed without modification alongside the unmodified execution report, source hash and checking code. Its 21,275 fixed-format rows have been checked a second time with an independent parser; all keys, mass numbers and mass excesses agree with the frozen DZ31 CSV (maximum printed difference 0.000 MeV). This verifies the DZ31 source-to-table values at printed precision, not its unavailable historical fitting roster or hypothetical unrounded numerical agreement. The standardized DZ31 values have RMS deviation 0.4093 MeV over 2339 primary AME2020 masses and 0.4267 MeV over 2457 broad, non-estimated AME2020 masses. The two different numbers reflect population definitions, not a verification against the same population in an external paper. Their scope is explicitly identified here.

## 4 Results

### *4.1 Spread–error discrimination and empirical coverage*

Table 3 reports the four population definitions. The primary development Spearman association is 0.422 ($p = 0.016$), and the primary confirmation association is 0.265 (nominal $p = 0.060$). Without the 100 keV cut, the correlations are 0.402 ($p = 0.0018$, $n = 58$) and 0.329 ($p = 0.0051$, $n = 71$) in development and confirmation. The broad-confirmation ordinary bootstrap 95% interval is $[0.092, 0.538]$; the isotopic-chain interval is $[0.055, 0.566]$. For primary confirmation, the corresponding intervals are $[-0.026, 0.521]$ and $[-0.057, 0.558]$. The positive broad result is not compatible with a blanket statement that spread contains no ordering information. Nevertheless, neither rank correlation by itself tests whether a quoted numerical uncertainty is calibrated.

The 20 excluded confirmation nuclei have mean error 0.446 MeV and mean spread 0.596 MeV; the 51 retained have mean error 0.247 and mean spread 0.435 MeV. One conspicuous case is $^{55}$Ca, with $E = 1.915$ MeV and $D = 1.013$ MeV, which the 100 keV uncertainty cut removes. Experimental uncertainty combined in quadrature with $D$ increases the broad number covered at $1\times$ the combined scale from 57/71 to 58/71. The exclusion does not make the omitted cases scientifically negligible.

Coverage is more directly interpretable than rank correlation for a proposed interval width: 42/51 primary and 57/71 broad confirmation errors lie within $\pm D$, while 49/51 and 68/71 lie within $\pm 2D$. These are empirical fractions, not nominal Bayesian credible levels. The idealized independent-four-model reference gives respective coverages about 86% and 97%, but substantial shared residual errors violate its premise. The 95% Wilson intervals for empirical coverage are $[0.697, 0.904]$ $(D)$ and $[0.868, 0.989]$ $(2D)$ for primary confirmation, and $[0.696, 0.879]$ and $[0.883, 0.986]$ for broad confirmation. The corresponding *isotopic-chain bootstrap* coverage intervals are $[0.690, 0.923]$ and $[0.894, 1.000]$ for primary $C(D)$ and $C(2D)$, respectively, and $[0.697, 0.890]$ and $[0.909, 1.000]$ for broad confirmation. The primary confirmation MAE is 0.247 MeV with a $Z$-block 95% interval $[0.192, 0.323]$ MeV; the broad MAE is 0.303 MeV with interval $[0.227, 0.403]$ MeV. The four-cohort block-interval table and the deterministic bootstrap CSV are supplied in the supplement. The full empirical distribution of $E/D$ is compared with the idealized $|t_3|/2$ law in Fig. 3; resemblance to this law does not prove its assumptions. The RMS scale ratio $k_{\mathrm{RMS}}$ changes from

**Table 3.** Independently reconstructed error ranking and scale diagnostics. Intervals on $\rho$ are 95% isotopic-chain bootstrap intervals. Coverage is the count of nuclei whose later absolute error is inside the stated band. $k_{\mathrm{RMS}}$ is defined in Eq. (2).

| Cohort | $n$ | $\rho$ | block CI | $E \leq D$ | $E \leq 2D$ | $k_{\mathrm{RMS}}$ |
|---|---|---|---|---|---|---|
| Development, primary | 32 | 0.422 | $[-0.005, 0.740]$ | 25 | 31 | 1.255 |
| Development, broad | 58 | 0.402 | $[0.042, 0.693]$ | 42 | 53 | 1.046 |
| Confirmation, primary | 51 | 0.265 | $[-0.057, 0.558]$ | 42 | 49 | 0.703 |
| Confirmation, broad | 71 | 0.329 | $[0.055, 0.566]$ | 57 | 68 | 0.818 |

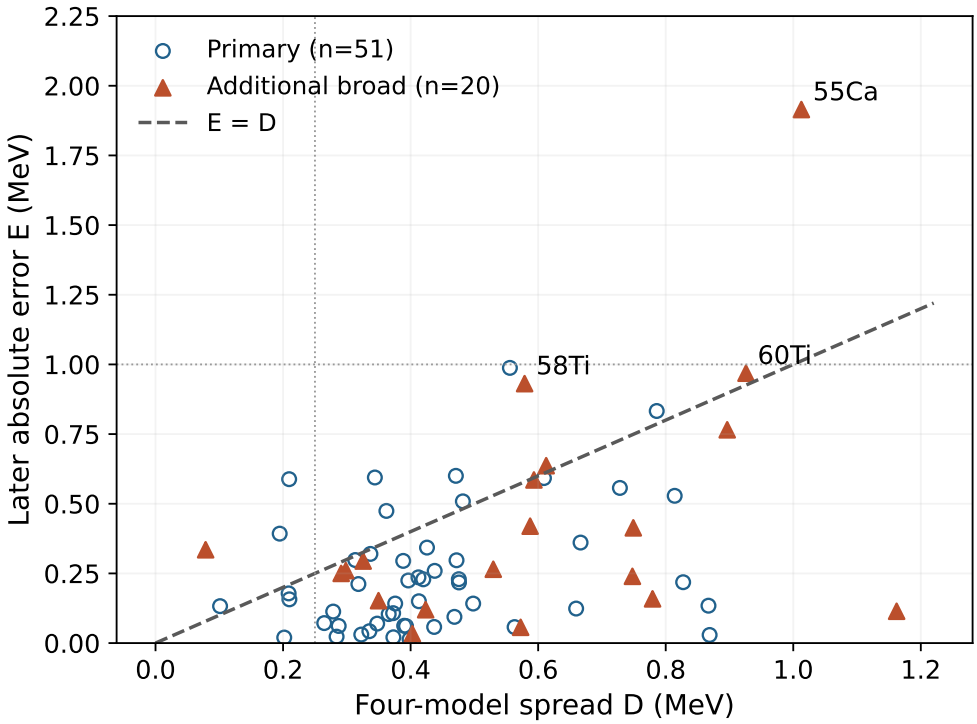



**Figure 2.** Later absolute mass-excess error versus four-model spread in the 71-nucleus confirmation population, distinguishing the 51 primary targets from 20 additional targets without the 100 keV experimental-uncertainty restriction. Open circles mark primary cases; filled triangles mark the additional cases. The diagonal is $E = D$; dotted horizontal and vertical lines show the absolute false-consensus thresholds $E = 1.0$ MeV and $D = 0.25$ MeV (FC-A). Points above the diagonal have error larger than the raw model spread.

1.255 in primary development to 0.703 in primary confirmation and 0.818 in broad confirmation. Isotopic-chain bootstrap intervals for these ratios are $[0.648, 1.478]$, $[0.537, 0.882]$ and $[0.593, 1.024]$, respectively. The numerical mapping is therefore sample-sensitive even though raw spread retains a positive aggregate association.

For the Normal predictive-score illustration, the development-scaled $k_{\mathrm{RMS,dev}}D$ yields mean CRPS 0.198 MeV on primary confirmation and 0.234 MeV on broad confirmation. A development-fitted constant-scale Normal benchmark gives 0.227 and 0.258 MeV, respectively. The spread-dependent working scale therefore improves this proper score on the same cases, yet is conservative when transferred: its nominal 90% Normal interval covers 50/51 and 69/71 confirmation masses. This is evidence of imperfect transfer of a *numerical* scale, not of an absence of useful discrimination. These exploratory scores should not be mistaken for calibrated model posteriors.

### *4.2 False consensus and predictive accuracy*

FC-A cannot discriminate a failure in the primary confirmation population: its largest error is only 0.988 MeV, so *no* nucleus reaches the event's 1 MeV error condition, independent of spread. The observed count 0/51 is therefore not a reassuring negative finding. Among the six targets having $D \leq 0.25$ MeV, the zero-event Wilson upper confidence bound is 0.390. For comparison, FC-A was observed in 1/32 development nuclei, as was FC-B. The broader confirmation set includes $^{55}$Ca above 1 MeV error, but its spread is well above the 0.25 MeV low-spread threshold, and FC-A remains 0/71. The separate FC-B rule identifies $^{30}$Cl and $^{77}$Cu (2/51, or 2/71 in broad confirmation). Both nuclei satisfy the frozen FC-B cutoffs, but they are not flagged throughout the full sensitivity

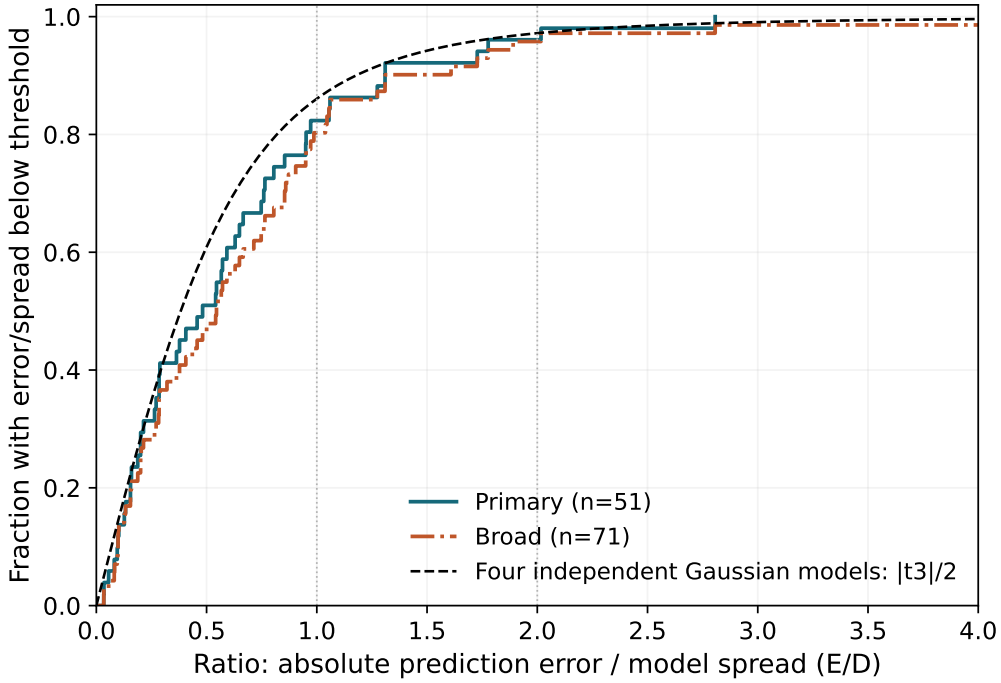



**Figure 3.** Empirical fraction of targets with absolute error/spread $E/D$ below each plotted value in the primary (solid teal, $n = 51$) and broad (dash-dotted orange, $n = 71$) confirmation cohorts. The dashed black curve is a *counterfactual* reference for four independent, identically distributed, zero-mean Gaussian model errors, where $E/D \sim |t_3|/2$; real-model dependence invalidates the reference as a guaranteed coverage law. Vertical dotted lines at ratios 1 and 2 indicate coverage within $\pm D$ and $\pm 2D$.

**Table 4.** Prediction accuracy (MAE, MeV). Means and medians refer to alternative centers of the four predictions, not to empirical uncertainty calibration.

| Sample | $n$ | $\rho(D, E)$ | mean MAE | median MAE |
|---|---|---|---|---|
| Development | 32 | 0.422 | 0.403 | 0.396 |
| Confirmation | 51 | 0.265 | 0.247 | 0.244 |

threshold ranges: $^{77}$Cu has $D = 0.210$ MeV, above 0.20 MeV, and $^{30}$Cl has $E = 0.393$ MeV, below 0.40 MeV. The quartile-derived cutoffs should not be regarded as physical constants.

$^{30}$Cl requires a special caveat. Its evaluated mass is constrained by proton-emission spectroscopy: Mukha *et al.* obtained the ground-state one-proton separation energy $S_p = -0.48(2)$ MeV from decay-in-flight correlations [23]. It was not a direct Penning-trap mass measurement. We retain the AME2020 non-estimated evaluated value in the predefined sample, but distinguish the experimental inference from direct mass spectrometry.

The four-model mean has confirmation MAE 0.247 MeV versus 0.559, 0.449, 0.368 and 0.224 MeV for FRDM2012, HFB-27, DZ31 and WS4, respectively. Thus WS4 by itself performs best on this limited set; combining one accurate member with three less accurate members is not necessarily advantageous. When FRDM2012, HFB-27, DZ31 or WS4 is removed in turn, the three-model consensus MAEs are 0.249, 0.261, 0.281 and 0.287 MeV. However, such leave-one-out changes alter both center and spread, and a stable MAE alone does not demonstrate stable calibration. The supplementary table therefore reports the corresponding three-model $D$–$E$ correlations and coverages. These correlations range from 0.164 when WS4 is omitted to 0.332 when DZ31 is omitted; model-membership robustness is therefore limited. In development, the four-model consensus MAE is 0.403 MeV, versus 0.653, 0.487, 0.647 and 0.283 MeV for the four members in the same order.

### *4.3 Chart conditions and shared residuals*

The prespecified shell-proximity comparison is a notable observation. The nine near-magic nuclei in primary confirmation have MAE 0.593 MeV, compared with 0.173 MeV for the other 42. Median $E/D$ is 1.060 versus 0.326. The near-minus-other MAE contrast is 0.420 MeV, with a 95% isotopic-chain bootstrap percentile interval $[0.274, 0.569]$ MeV, calculated by recomputing each subgroup mean within the resampled chains. The broad-cohort contrast is 0.413 MeV with interval $[0.213, 0.631]$ MeV. These are descriptive block-bootstrap intervals, not proof that shell structure

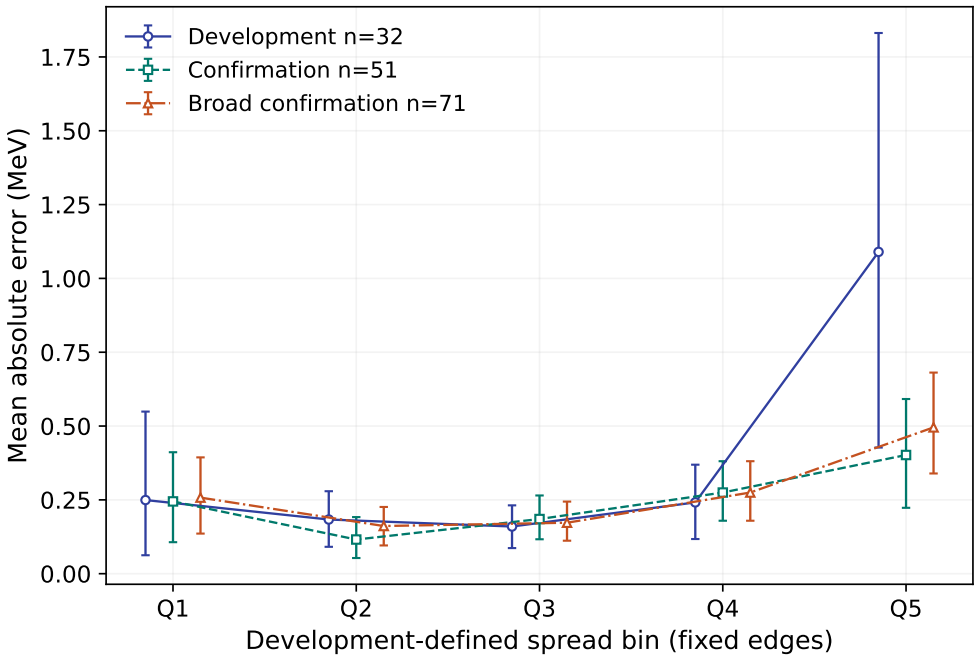



**Figure 4.** Mean later error in spread bins whose cut points were defined on the 32 primary development nuclei. Bars are ordinary 95% bootstrap intervals (10,000 resamples; seed 42); the 68% intervals and bin counts are tabulated in the Supplement. Circles denote primary development ($n = 32$), squares primary confirmation ($n = 51$), and triangles broad confirmation ($n = 71$). Apparent nonmonotonicity of the point estimates is uncertain with such small fixed-bin populations.

causes the excess error. A descriptive least-squares fit

$$E_i = 0.111 + 0.150 D_i + 0.404\,\mathbf{1}_{\text{near shell},i} \quad \text{(MeV)} \tag{4}$$

has heteroskedasticity-robust (HC3) standard error about 0.086 MeV for the shell-indicator coefficient. HC3 corrects heteroskedasticity at the nucleus level and is not a chain-clustered standard error; the separate chain-bootstrap MAE contrast above addresses a different descriptive quantity. Removing each near-magic nucleus in turn leaves that coefficient between 0.356 and 0.464 MeV. The association does not disappear upon deletion of one flagged case, but nine correlated cases cannot distinguish a shell mechanism from unmeasured regional confounders. Specifically, $^{101}$In is flagged for $Z = 49$ and $N = 52$ close to 50; $^{220}$Np is flagged because $N = 127$ is near the 126 neutron shell.

Extrapolation distance is too restricted to be a strong explanation on its own. Every primary confirmation nucleus is at $d_1 \leq 3$ and $d_2 \leq \sqrt{5}$ from a primary measured earlier-vintage nucleus; the broad population extends only to $d_1 = 4$ and $d_2 = \sqrt{13}$. The primary Spearman association between $d_2$ and error is 0.048 ($p = 0.737$); in broad confirmation it is 0.170 ($p = 0.157$). Only 7/51 primary confirmation targets have $Z \leq 30$; 19 are in the neutron-rich rare-earth cluster $59 \leq Z \leq 65$, $97 \leq N \leq 103$, and 15 have $Z \geq 91$. There are only 30 represented isotopic chains. Thus this is largely a short-range interpolation/near-boundary check, not a test in the remote $r$-process chart.

Pairwise correlations among signed model residuals range from 0.504 to 0.837 in development and from −0.049 to 0.576 in primary confirmation. Different chart compositions (notably neutron-rich lighter nuclei versus rare earths and actinides) are plausible contributors, but the sample does not identify the cause. Dependence prevents treating $D/2$ as a four-draw standard error.

### *4.4 Two-neutron separation energy*

There are 34 primary confirmation targets with non-estimated measured $N - 2$ neighbors. Their consensus $S_{2n}$ MAE is 0.126 MeV and their spread–error Spearman coefficient is 0.302 ($p = 0.083$). In the broad population, 44 qualify, with $S_{2n}$ MAE 0.182 MeV. For the 34-case primary population, the individual $S_{2n}$ MAEs are 0.158, 0.186, 0.240 and 0.126 MeV for FRDM2012, HFB-27, DZ31 and WS4, respectively. Subtraction cancels smoothly varying errors, so smaller separation-energy errors do not imply that mass-excess uncertainties or probabilities are well calibrated. Since adjacent masses also share experimental and theoretical error structure, the 34 separation-energy cases are not independent evidence for mass calibration.

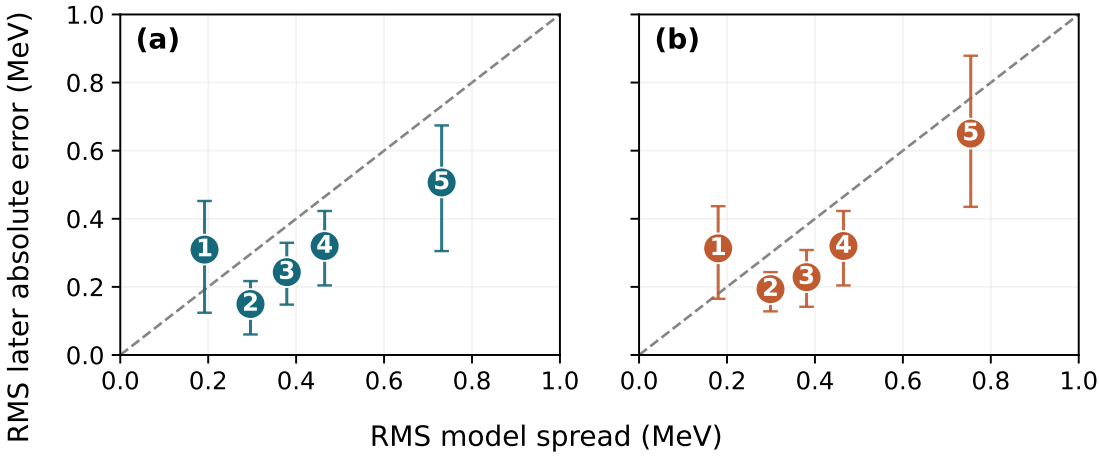



**Figure 5.** Magnitude diagnostic complementary to rank correlation, with (a) the 51-nucleus primary confirmation cohort and (b) the 71-nucleus broad sensitivity cohort. Each numbered circle identifies one of the five development-defined spread bins (1–5 denotes Q1–Q5) and plots the root mean square (RMS) of later absolute errors against the RMS four-model spread, both in MeV. In bin order Q1–Q5, the counts are 6, 7, 17, 10, 11 in (a) and 7, 10, 20, 10, 24 in (b). Vertical bars show ordinary 95% bootstrap intervals for RMS error. The dashed equality line is a diagnostic guide rather than a universal calibration law.

### *4.5 Calcium-chain comparison with regional shell-model studies*

The analysis plan specified the calcium chain $Z = 20$, $28 \leq N \leq 40$ as RQ3, with Li *et al.*'s realistic Gamow shell model (GSM) and Coraggio *et al.*'s chiral-force shell model as regional structural references [21, 22]. The original data-source rule restricted numerical comparisons to tables, author data or reproducible calculations. A documented amendment made during revision admits values reconstructed from the vector graphics of Li *et al.*'s Figure 2(b), after requests for the original numerical outputs remained unanswered. RQ3 was planned; this data-acquisition method was adopted during revision after the figure-derived values had been inspected. The frozen protocol is preserved, and the amendment is supplied with the analysis. The GSM series is not an additional global-ensemble member and changes none of the primary mass-error results.

Figure 7 compares the four global models, the published Coraggio Table II values, and the reconstructed GSM series against the same AME2020 $S_{2n}$ references. Both masses must be non-estimated for an experimental point to enter this comparison. The GSM $^{48}$Ca-core and $^{54}$Ca-core calculations are retained separately, including their overlap at $A = 56, 58, 60$. There is no GSM $S_{2n}$ point for $^{48}$Ca in the source figure, and none is imputed. AME extrapolations at $A = 58, 60$ are not scored as measurements. Values through $A = 72$ are archived in Supplement S7 for completeness; only $A \leq 60$ enters the planned regional comparison.

The primary GSM source is the published article by Li *et al.*, Phys. Rev. C **102**, 034302 (2020), Figure 2(b) [21]. All 15 GSM entries agree with its vector coordinates when rounded to 0.01 MeV. The maximum rounded-value discrepancy from the published figure is 0.00461 MeV. These are checks of figure reconstruction, not recovery of the original solver output or estimates of theoretical uncertainty. Supplement S7 reports the calibration diagnostics and sensitivity to adopted extraction allowances up to 0.10 MeV.

The previously tabulated five-isotope comparison ($A = 48, 50, 52, 54, 56$) is retained in Supplement S7. For a comparison including GSM, all six methods are instead scored on their common measured subset $A = 50, 52, 54, 56$. The mean absolute $S_{2n}$ errors are 1.159, 0.795, 0.914 and 0.220 MeV for FRDM2012, HFB-27, DZ31 and WS4, 0.680 MeV for Coraggio, and 0.755 MeV for GSM with the $^{48}$Ca core. Allowing each GSM value to vary independently by an adopted ±0.10 MeV changes its MAE to a possible range of 0.655–0.855 MeV. This stress-test range is not a confidence interval or a proven error bound on the underlying calculation; it also shows that the small Coraggio–GSM MAE difference is not a robust ranking.

At $^{56}$Ca, AME2020 gives $S_{2n} = 4.492$ MeV, with an approximately propagated uncertainty of 0.254 MeV. FRDM2012, HFB-27, DZ31 and WS4 give 6.153, 5.943, 5.593 and 4.482 MeV, Coraggio gives 3.377 MeV, and the two GSM core choices give 3.95 and 4.37 MeV. Their GSM absolute residuals are therefore 0.542 and 0.122 MeV. The $^{54}$Ca-core result has only this one measured overlap in the selected region; it is not substituted into the four-point $^{48}$Ca-core series. The approximately 250-keV mass uncertainty of $^{56}$Ca excludes it from the primary 51-nucleus mass cohort, while it remains eligible for this broader regional comparison.

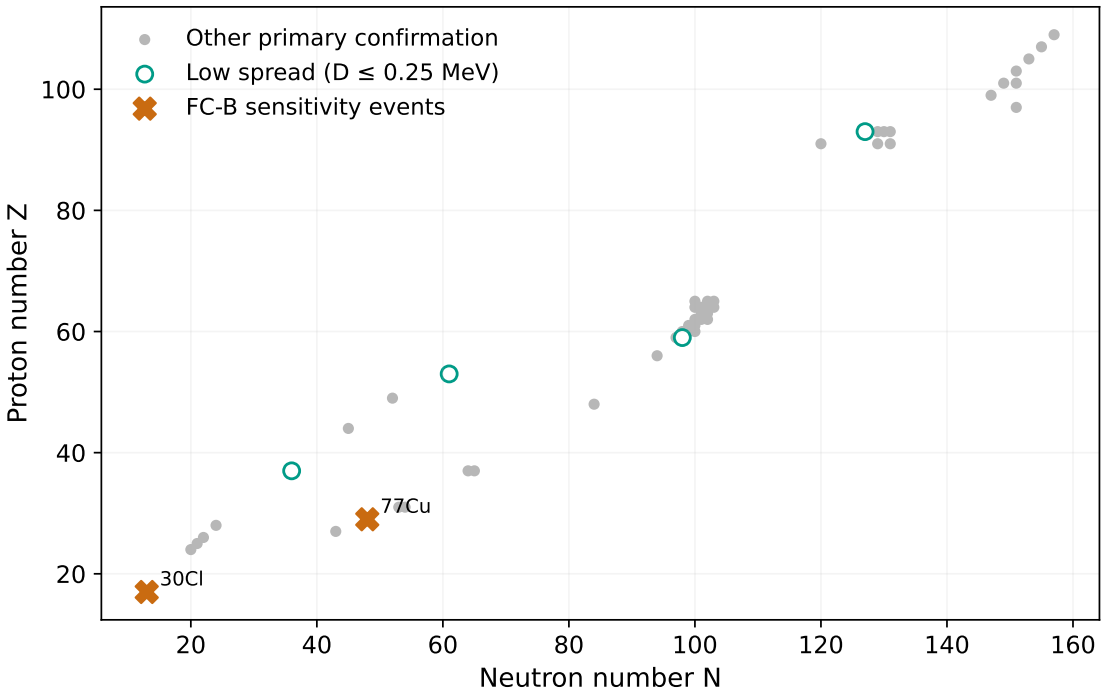



**Figure 6.** Locations of low-spread targets among the 51 primary confirmation nuclei on the neutron–proton $(N, Z)$ chart. Gray points are the other targets, open teal circles have $D \leq 0.25$ MeV, and orange crosses identify $^{30}$Cl and $^{77}$Cu, which satisfy the less stringent fixed FC-B definition ($D \leq 0.238$ MeV, $E \geq 0.341$ MeV). No FC-A event ($D \leq 0.25$ MeV, $E \geq 1.0$ MeV) occurs in this population.

The GSM $^{48}$Ca-core decreases $S_{2n}(52) - S_{2n}(54) = 4.12$ MeV and $S_{2n}(54) - S_{2n}(56) = 4.07$ MeV can be compared with the AME2020 decreases 3.783 and 2.545 MeV. The calculation represents the declines across $N = 32$ and 34 but overestimates the latter. Thus the regional calculations supply structural information without demonstrating uniformly smaller errors than the global models. Both 2020 shell-model comparisons are retrospective; Li *et al.* used an experimental $^{55}$Ca separation energy to set a single-particle potential [21]. The small overlap and this calibration history preclude an independent test that shell-model physics resolves apparent false consensus.

## 5 Discussion

A positive rank association is seen in both confirmation populations, and the empirical $\pm D$ and $\pm 2D$ intervals cover substantial fractions of the later evaluated masses. Those observations cannot be summarized as "agreement is not calibrated" solely because the 51-case $p$-value exceeds 0.05. Conversely, the roughly 80% and 96% empirical coverages do not certify nominal 68% and 95% frequentist bands: $D$ is computed from four dependent deterministic predictions with unequal accuracies, not four exchangeable error draws.

The scale ratio and CRPS comparisons quantify the sample dependence of this relationship. Spread offers some information relative to a fixed-width benchmark in this restricted sample, while an error-scale transformation estimated in one vintage is conservative on the next. The nonmonotonic quintile point estimates come with broad intervals and should not be interpreted as a conclusive failure of monotonicity. WS4's superior individual performance is another warning against treating an unweighted heterogeneous mean as optimal. These qualifications are consistent with earlier Bayesian calibration and model-mixing studies [6, 17, 18, 20] rather than contradicting them.

There is a physically relevant shell-proximity association, but an observational shell flag is not a mechanism. Bulk differences between models may reflect macroscopic terms, whereas local structure can shift individual masses and separation energies, as explored by Ye and Wan and Giuliani *et al.* [5, 7]. That interpretation motivates, but does not establish, the calcium mechanism. The two regional calculations provide quantitative comparisons, but their small retrospective overlap does not identify a causal mechanism. A larger, independent sample remains necessary before claiming that shell-model physics resolves apparent false consensus.

The limits of inference are substantial. The main cohort consists of only 51 nuclei from 30 isotopic chains with concentrated rare-earth and actinide representation; simple i.i.d. intervals can understate uncertainty. All primary confirmation targets are within at most three $L^1$ lattice steps of

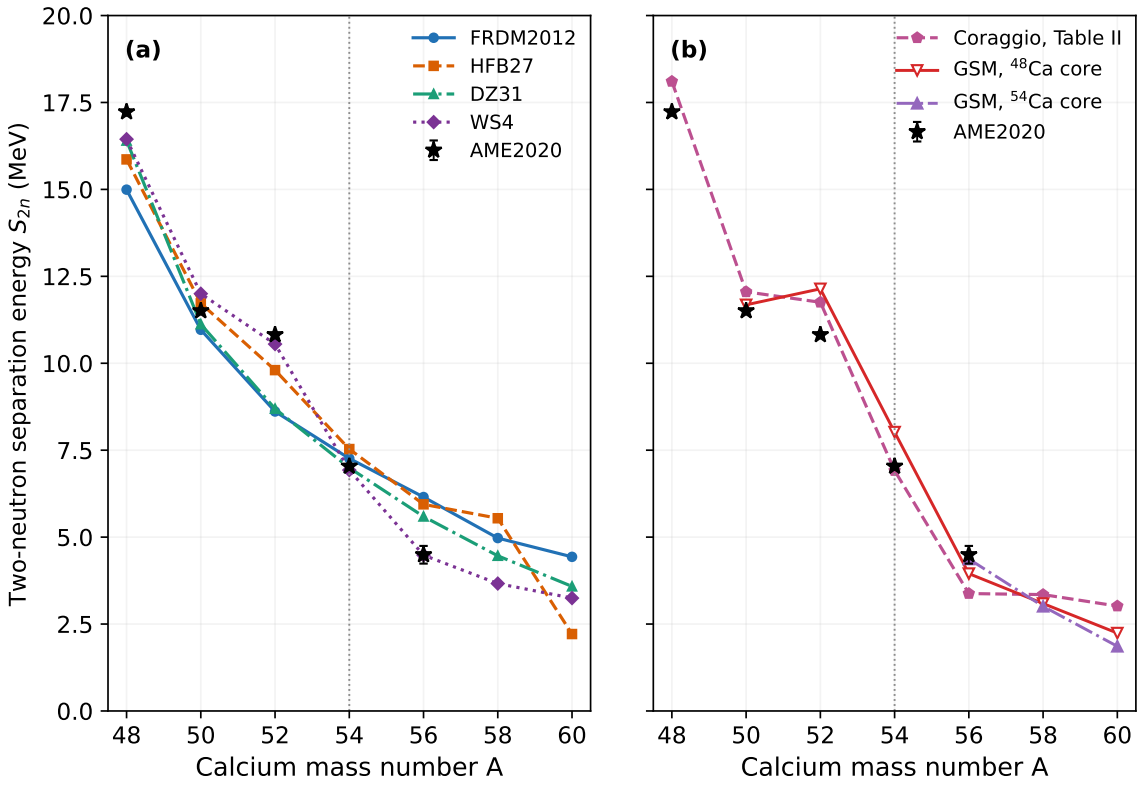



**Figure 7.** Calcium $S_{2n}$ for $A = 48$–60. (a) Four global mass models. (b) Coraggio *et al.*'s published Table II and the two GSM core choices reconstructed from Li *et al.*'s Figure 2(b). Black stars in both panels are the same AME2020 values with non-estimated input masses. Experimental bars use quadrature of the two mass uncertainties; unavailable covariance makes these bars approximate. Downward open triangles denote the GSM $^{48}$Ca-core series; upward triangles denote the $^{54}$Ca-core series. Both are figure-derived estimates, with extraction sensitivity reported in Supplement S7; no theoretical uncertainty bars are inferred. The dotted line marks $^{54}$Ca ($N = 34$). Legends are in the upper right of each panel, and lines connect calculated points only.

a mass already evaluated in AME2016. The experiment is therefore not a validation of model uncertainty for remote $r$-process nuclides. The 100 keV filter changes which nuclear regions appear in the sample, some experimentally evaluated masses are derived indirectly from decay data, and the fit datasets of these four models are not identically timed. The source audit independently checks all published AME rows and the available RIPL-4 tables. The original BRUSLIB HFB-27 conversion remains unavailable, although the DZ31 executable has now been checked against all archived predictions at their printed precision.

## 6 Conclusion

On later AME evaluations, the fixed, equally weighted FRDM2012/HFB-27/DZ31/WS4 spread positively ranks error in both the 51-nucleus primary sample ($\rho = 0.265$) and the prespecified 71-nucleus sensitivity sample ($\rho = 0.329$). Approximately four fifths of errors fall within $\pm D$ and 96% within $\pm 2D$ in either confirmation population, although these bands have no automatic nominal confidence interpretation. The mapping from spread to numerical error changes across the historical cohorts, and near-magic cases have larger average errors not fully described by the spread. These are useful local findings, not a universal calibration or an extrapolation test far from experimentally known nuclei. For nuclear applications, model-disagreement bands should be reported as such and quantitatively validated for the model set, observable, region and measurement chronology before probabilistic meaning is assigned.

## Acknowledgements

The author acknowledges Palestine Polytechnic University (PPU) for its institutional support. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

**Data availability statement**
The published baseline repository location is:
https://github.com/ozalloum/nuclear-mass-model-consensus-calibration/releases/tag/v1.2.1

**ORCID iDs**
Othman H. Y. Zalloum 0000-0001-7282-1332

**References**

[1] P. Möller, A. J. Sierk, T. Ichikawa, and H. Sagawa, Nuclear ground-state masses and deformations: FRDM(2012), At. Data Nucl. Data Tables **109–110**, 1 (2016), https://doi.org/10.1016/j.adt.2015.10.002.

[2] S. Goriely, N. Chamel, and J. M. Pearson, Hartree–Fock–Bogoliubov nuclear mass model with 0.50 MeV accuracy based on standard forms of Skyrme and pairing functionals, Phys. Rev. C **88**, 061302(R) (2013), https://doi.org/10.1103/PhysRevC.88.061302.

[3] J. Duflo and A. P. Zuker, Microscopic mass formulas, Phys. Rev. C **52**, R23 (1995), https://doi.org/10.1103/PhysRevC.52.R23.

[4] N. Wang, M. Liu, X. Wu, and J. Meng, Surface diffuseness correction in global mass formula, Phys. Lett. B **734**, 215 (2014), https://doi.org/10.1016/j.physletb.2014.05.049.

[5] S. A. Giuliani *et al.*, Impact of nuclear masses on r-process nucleosynthesis: Bulk properties versus shell effects, Phys. Rev. C **113**, 045806 (2026), https://doi.org/10.1103/k64z-32sr.

[6] Y. Saito *et al.*, Uncertainty quantification of mass models using ensemble Bayesian model averaging, Phys. Rev. C **109**, 054301 (2024), https://doi.org/10.1103/PhysRevC.109.054301.

[7] W. Ye and N. Wan, Understanding on prediction differences among theoretical mass models with machine learning techniques, Phys. Rev. C **111**, 044317 (2025), https://doi.org/10.1103/PhysRevC.111.044317.

[8] M.-H. Chen *et al.*, Impact of nuclear mass models on r-process nucleosynthesis and heavy element abundances in r-process-enhanced metal-poor stars, Astron. Astrophys. **693**, A1 (2025), https://doi.org/10.1051/0004-6361/202451986.

[9] G. Audi, M. Wang, A. H. Wapstra, F. G. Kondev, M. MacCormick, X. Xu, and B. Pfeiffer, The AME2012 atomic mass evaluation (I). Evaluation of input data, adjustment procedures, Chin. Phys. C **36**, 1287 (2012), https://doi.org/10.1088/1674-1137/36/12/002.

[10] M. Wang, G. Audi, A. H. Wapstra, F. G. Kondev, M. MacCormick, X. Xu, and B. Pfeiffer, The AME2012 atomic mass evaluation (II). Tables, graphs and references, Chin. Phys. C **36**, 1603 (2012), https://doi.org/10.1088/1674-1137/36/12/003.

[11] M. Wang *et al.*, The AME2016 atomic mass evaluation (II). Tables, graphs and references, Chin. Phys. C **41**, 030003 (2017), https://doi.org/10.1088/1674-1137/41/3/030003.

[12] W. J. Huang, G. Audi, M. Wang, F. G. Kondev, S. Naimi, and X. Xu, The AME2016 atomic mass evaluation (I). Evaluation of input data and adjustment procedures, Chin. Phys. C **41**, 030002 (2017), https://doi.org/10.1088/1674-1137/41/3/030002.

[13] W. J. Huang, M. Wang, F. G. Kondev, G. Audi, and S. Naimi, The AME2020 atomic mass evaluation (I). Evaluation of input data and adjustment procedures, Chin. Phys. C **45**, 030002 (2021), https://doi.org/10.1088/1674-1137/abddb0.

[14] M. Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi, The AME2020 atomic mass evaluation (II). Tables, graphs and references, Chin. Phys. C **45**, 030003 (2021), https://doi.org/10.1088/1674-1137/abddaf.

[15] Atomic Mass Data Center, *Duflo–Zuker mass formula software package*, including the 31-parameter executable, https://www-nds.iaea.org/amdc/.

[16] R. Utama and J. Piekarewicz, Validating neural-network refinements of nuclear mass models, Phys. Rev. C **97**, 014306 (2018), https://doi.org/10.1103/PhysRevC.97.014306.

[17] L. Neufcourt, Y. Cao, W. Nazarewicz, and F. Viens, Bayesian approach to model-based extrapolation of nuclear observables, Phys. Rev. C **98**, 034318 (2018), https://doi.org/10.1103/PhysRevC.98.034318.

[18] V. Kejzlar, L. Neufcourt, W. Nazarewicz, and P.-G. Reinhard, Statistical aspects of nuclear mass models, J. Phys. G **47**, 094001 (2020), https://doi.org/10.1088/1361-6471/ab907c.

[19] D. R. Phillips *et al.*, Get on the BAND Wagon: a Bayesian framework for quantifying model uncertainties in nuclear dynamics, J. Phys. G **48**, 072001 (2021), https://doi.org/10.1088/1361-6471/abf1df.

[20] P. Giuliani, K. Godbey, V. Kejzlar, and W. Nazarewicz, Model orthogonalization and Bayesian forecast mixing via principal component analysis, Phys. Rev. Research **6**, 033266 (2024), https://doi.org/10.1103/PhysRevResearch.6.033266.

[21] J. G. Li, B. S. Hu, Q. Wu, Y. Gao, S. J. Dai, and F. R. Xu, Neutron-rich calcium isotopes within realistic Gamow shell model calculations with continuum coupling, Phys. Rev. C **102**, 034302 (2020), https://doi.org/10.1103/PhysRevC.102.034302.

[22] L. Coraggio, G. De Gregorio, A. Gargano, N. Itaco, T. Fukui, Y. Z. Ma, and F. R. Xu, Shell-model study of calcium isotopes toward their drip line, Phys. Rev. C **102**, 054326 (2020), https://doi.org/10.1103/PhysRevC.102.054326.

[23] I. Mukha *et al.*, Deep excursion beyond the proton dripline. I. Argon and chlorine isotope chains, Phys. Rev. C **98**, 064308 (2018), https://doi.org/10.1103/PhysRevC.98.064308.